

Evaluating Human-Robot Skill Gaps in Electrical Circuit Inspection: A New Electronic Task Board for Benchmarking Manipulation

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ABSTRACT

This paper introduces a standardized benchmarking board to evaluate human vs. robot manipulation skills in delicate electronic assembly and circuit inspection. It analyzes force limits, coordinate tracking accuracy, and repeatability under different control frameworks.

RELEVANCE TO THE EB15 ROBOTIC ARM PROJECT

Provides methodologies for experimental validation and coordinate tracking repeatability tests of the EB15.

TECHNICAL ANALYSIS & SYSTEM INTEGRATION

This academic research reference documents crucial design paradigms directly relevant to the development of the EB15 six-degrees-of-freedom robotic manipulator. Under the distributed control framework designed for this thesis (featuring the ESP32 S3 as master and the Arduino Uno as slave), the mathematical formulations, communication latencies, and performance profiles analyzed by Peter So provide a benchmark for physical and algorithmic modeling. Specifically, this reference establishes solid validation criteria for timing constraints, closed-loop feedback via absolute magnetic encoders (AS5600), and vibration damping.